MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

BOMB, GENERAL PURPOSE, MK 81 MOD 0 (250LB)
IN PALLET, BOMB, MK 8 MOD 0
FLEET ISSUE UNIT LOAD

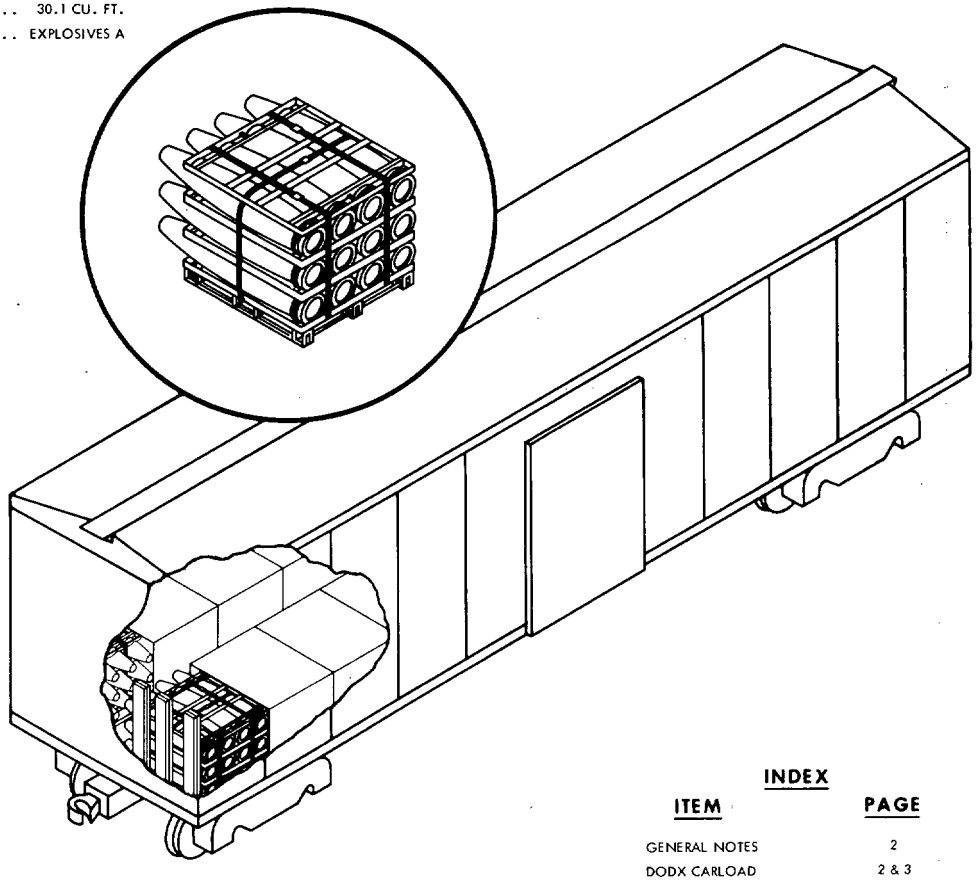
MIL-STD-1325-9
(NAVY)

9 NOVEMBER 1973

SUPERSEDING
WR-52/9
7 APRIL 1966

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/15
WEIGHT (APPROXIMATE) ..... 2978 LBS.
DIMENSIONS ... 44 1/2 L x 36 3/8 W x 32 1/8 H
CUBE ..... 30.1 CU. FT.
DOT HAZARD CLASSIFICATION .. EXPLOSIVES A



INDEX

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DODX CARLOAD	2 & 3
40' 6" CARLOAD COM'L	4 & 5
50' 6" CARLOAD COM'L	6 & 7

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

H.S. Gray NWHL 11/8/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 11/8/73
SIGNATURE (AIR-SYSCOM, BY DIRECTION) DATE

APPROVED BY BUREAU OF EXPLOSIVES

MILITARY ASSISTANT
R.J. Grassmuck 10/25/73
SIGNATURE DATE

ORIGINATOR Charles McBride 11/6/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

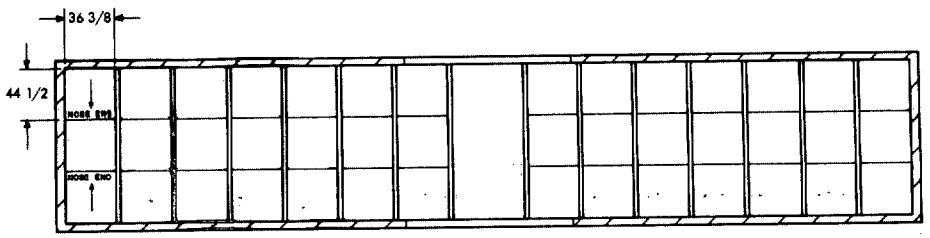
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD-LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF 250 LB MK 81 MOD 0 GENERAL PURPOSE BOMBS PALLETIZED IN ACCORDANCE WITH WR-54/15.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC MM-L-751
    NAILS - FED. SPEC FF-N-105
    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED)
    SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL 1).

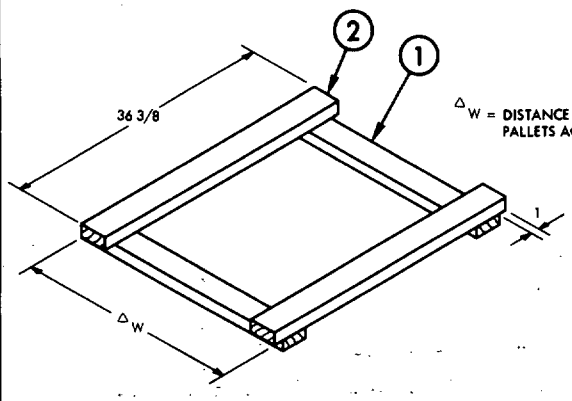
50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 28 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2200 LBS OF THE LADING.

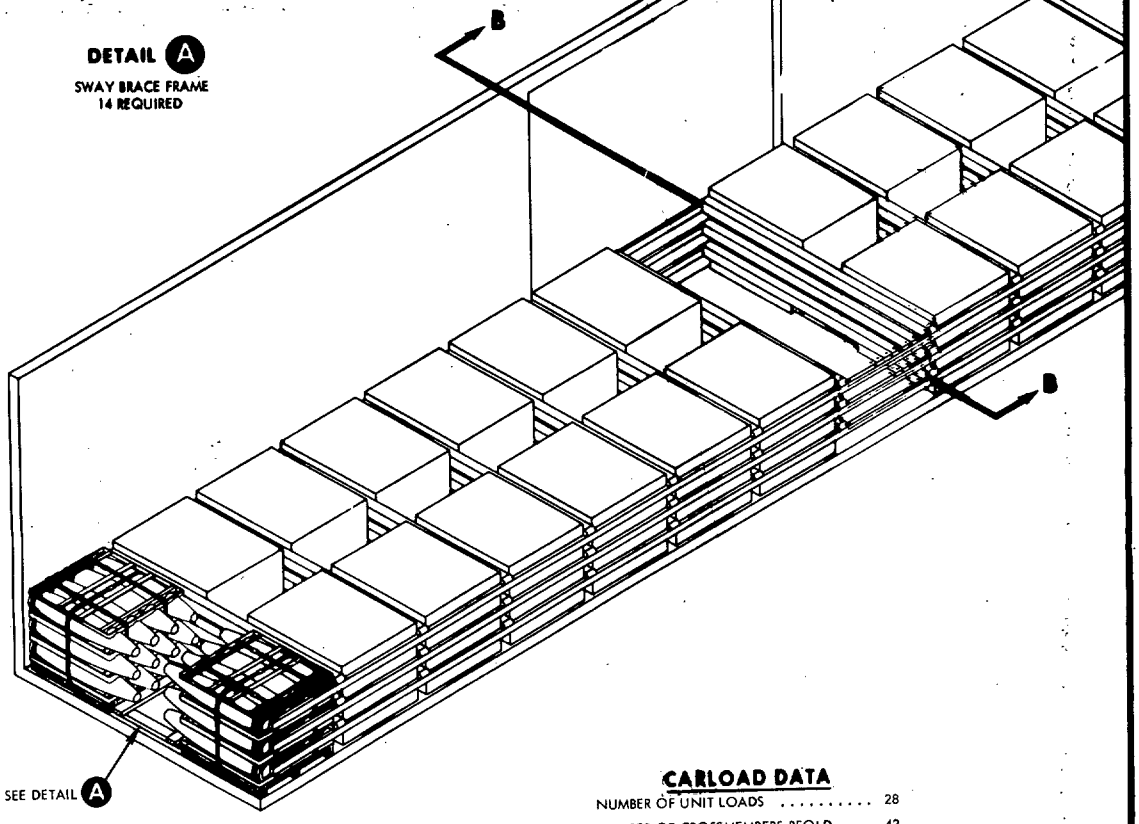


LOADING PLAN

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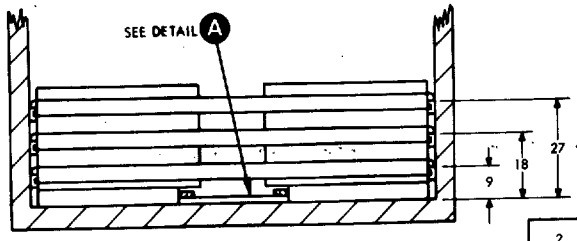


DETAIL A
SWAY BRACE FRAME
14 REQUIRED



CARLOAD DATA

NUMBER OF UNIT LOADS	28
NUMBER OF CROSSMEMBERS REQ'D	42
LOAD WEIGHT (APPROXIMATE)	83,384 LBS
DUNNAGE WEIGHT (APPROXIMATE)	271 LBS
CARLOAD WEIGHT (APPROXIMATE)	83,655 LBS



SECTION B-B
SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSS MEMBERS

* NAIL LATERAL MEMBERS OF SWAY BRACE TO CAR FLOOR WITH TWO 16d NAILS EACH END.

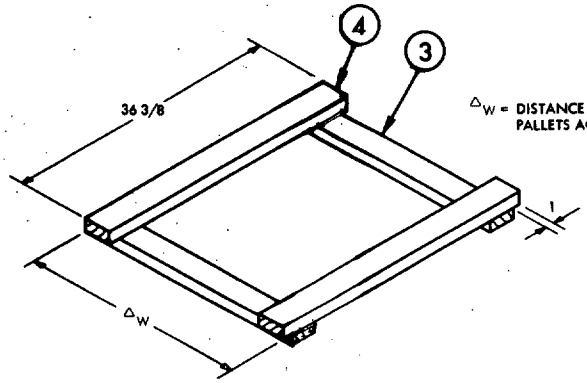
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
2	SWAY BRACE LONGITUDINAL	2 x 4 x 36 3/8	28	1	3 PER JOINT	10d
1	SWAY BRACE LATERAL	2 x 4 x CUT TO FIT	28	CAR FLOOR	SEE NOTE *	

LIST OF MATERIALS AND NAILING DATA

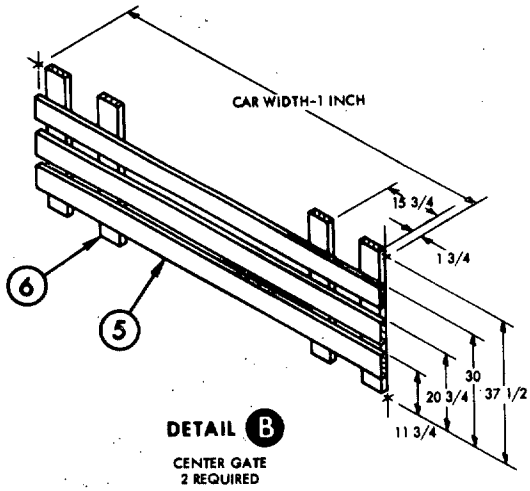
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40 FT 6 IN. BOXCAR, COMMERCIAL

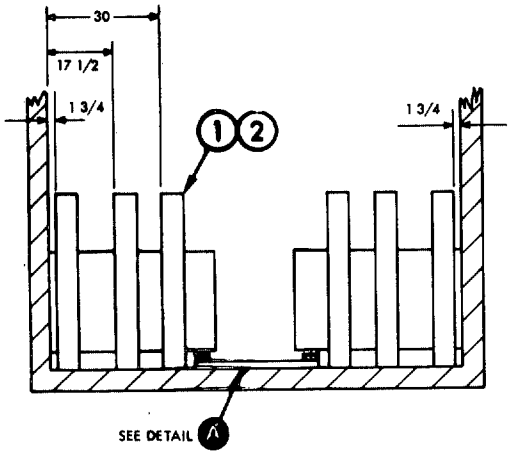
1. THE CARLOAD CONSISTS OF 26 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE PROVIDED DOORWAY PROTECTION WHEN REQUIRED IS INSTALLED IN ACCORDANCE WITH MIL-STD-1325 (NAVY).



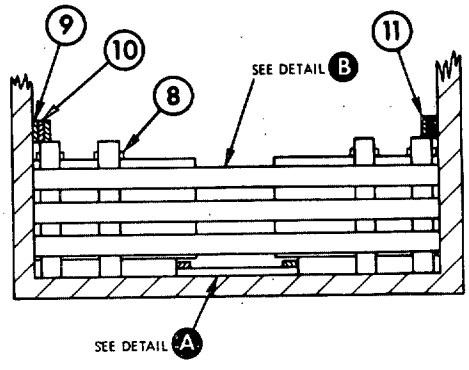
DETAIL A
SWAY BRACE FRAME
13 REQUIRED



DETAIL B
CENTER GATE
2 REQUIRED

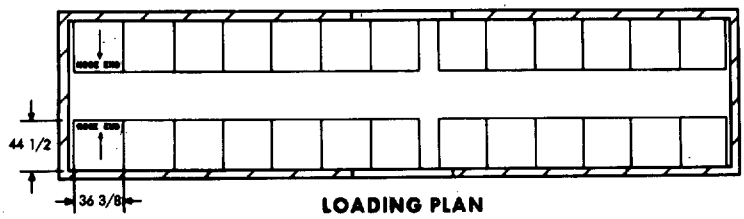


SECTION C-C
SHOWING LOCATION OF END WALL MEMBERS



SECTION D-D

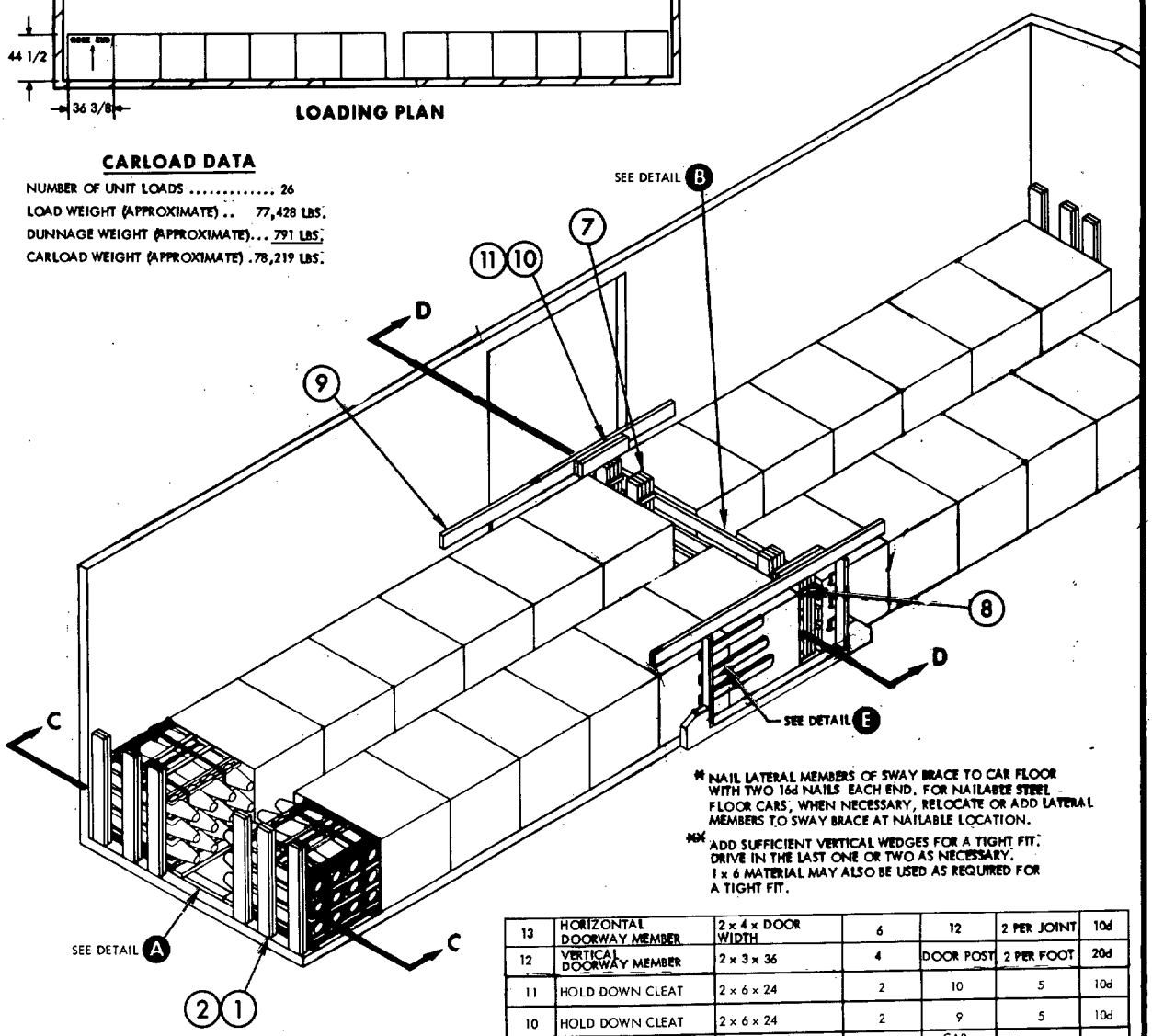
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LOADING PLAN

CARLOAD DATA

NUMBER OF UNIT LOADS 26
LOAD WEIGHT (APPROXIMATE) .. 77,428 LBS.
DUNNAGE WEIGHT (APPROXIMATE)... 791 LBS.
CARLOAD WEIGHT (APPROXIMATE) .78,219 LBS.

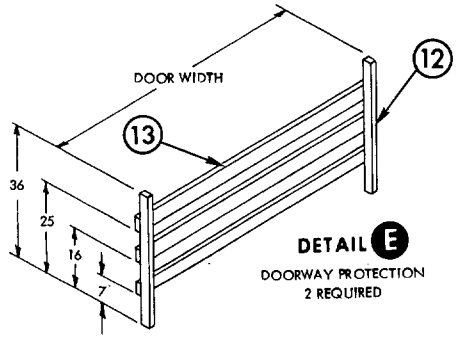


* NAIL LATERAL MEMBERS OF SWAY BRACE TO CAR FLOOR WITH TWO 16d NAILS EACH END. FOR NAILABLE STEEL FLOOR CARS, WHEN NECESSARY, RELOCATE OR ADD LATERAL MEMBERS TO SWAY BRACE AT NAILABLE LOCATION.

** ADD SUFFICIENT VERTICAL WEDGES FOR A TIGHT FIT. DRIVE IN THE LAST ONE OR TWO AS NECESSARY. 1 x 6 MATERIAL MAY ALSO BE USED AS REQUIRED FOR A TIGHT FIT.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
LIST OF MATERIALS AND NAILING DATA						
13	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	6	12	2 PER JOINT	10d
12	VERTICAL DOORWAY MEMBER	2 x 3 x 36	4	DOOR POST	2 PER FOOT	20d
11	HOLD DOWN CLEAT	2 x 6 x 24	2	10	5	10d
10	HOLD DOWN CLEAT	2 x 6 x 24	2	9	5	10d
9	GATE HOLD DOWN	2 x 6 x DOOR WIDTH + 48	2	CAR WALL	5 EACH END	10d
8	CENTER GATE TIE CLEAT	1 x 2 x TO SUIT	8	6,7	1 PER JOINT	10d
7	CENTER GATE VERTICAL WEDGES	2 x 6 x 37 1/2	AS REQ'D	SEE NOTE **		
6	CENTER GATE VERTICAL	2 x 6 x 37 1/2	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	6	6	3 PER JOINT	10d
4	SWAY BRACE LONGITUDINAL	2 x 4 x 36 3/8	26	3	3 PER JOINT	10d
3	SWAY BRACE LATERAL	2 x 4 x CUT TO FIT	26	CAR FLOOR	SEE NOTE *	
2	END WALL MEMBER	2 x 6 x 48	12	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 48	12	CAR WALL	1 PER FOOT	10d

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

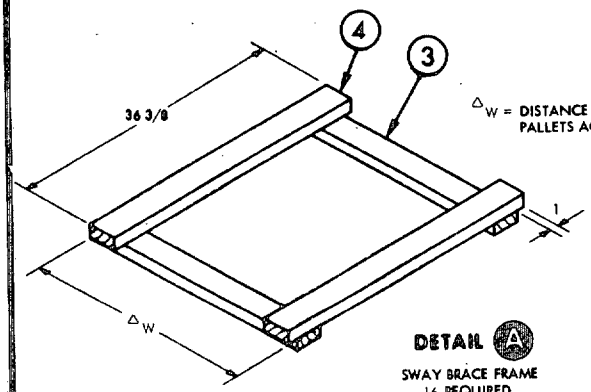


DETAIL E
DOORWAY PROTECTION
2 REQUIRED

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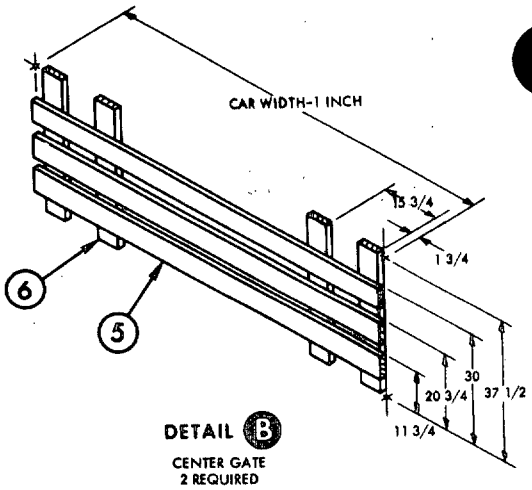
50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 32 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN-HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK (SEE DETAIL E) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DOORWAY PROTECTION, **DETAIL F**, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.

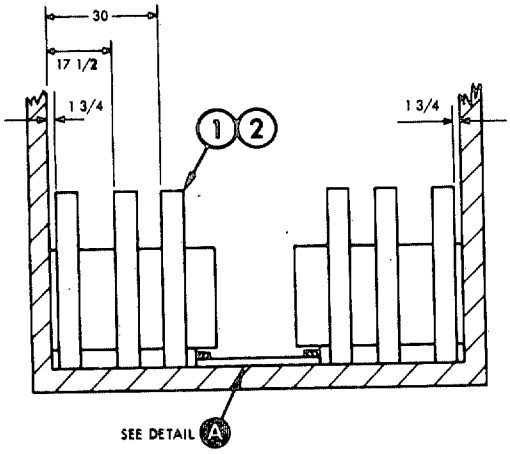


△W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR.

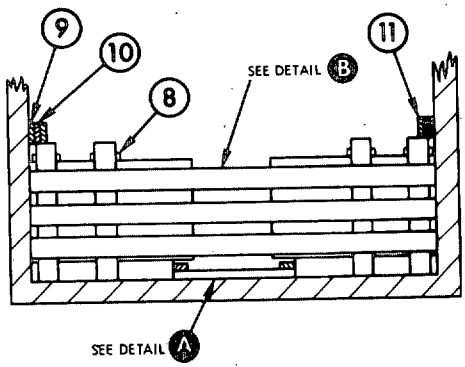
DETAIL A
SWAY BRACE FRAME
16 REQUIRED



DETAIL B
CENTER GATE
2 REQUIRED

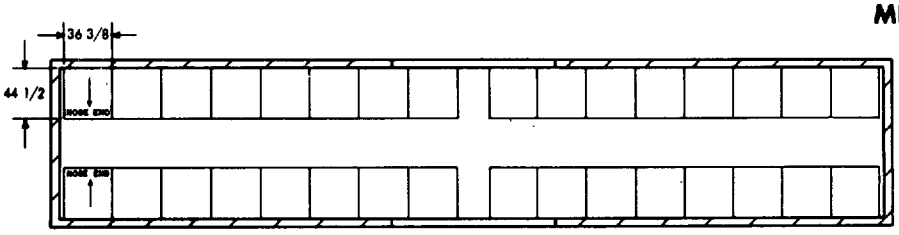


SECTION C-C
SHOWING LOCATION OF END WALL MEMBERS

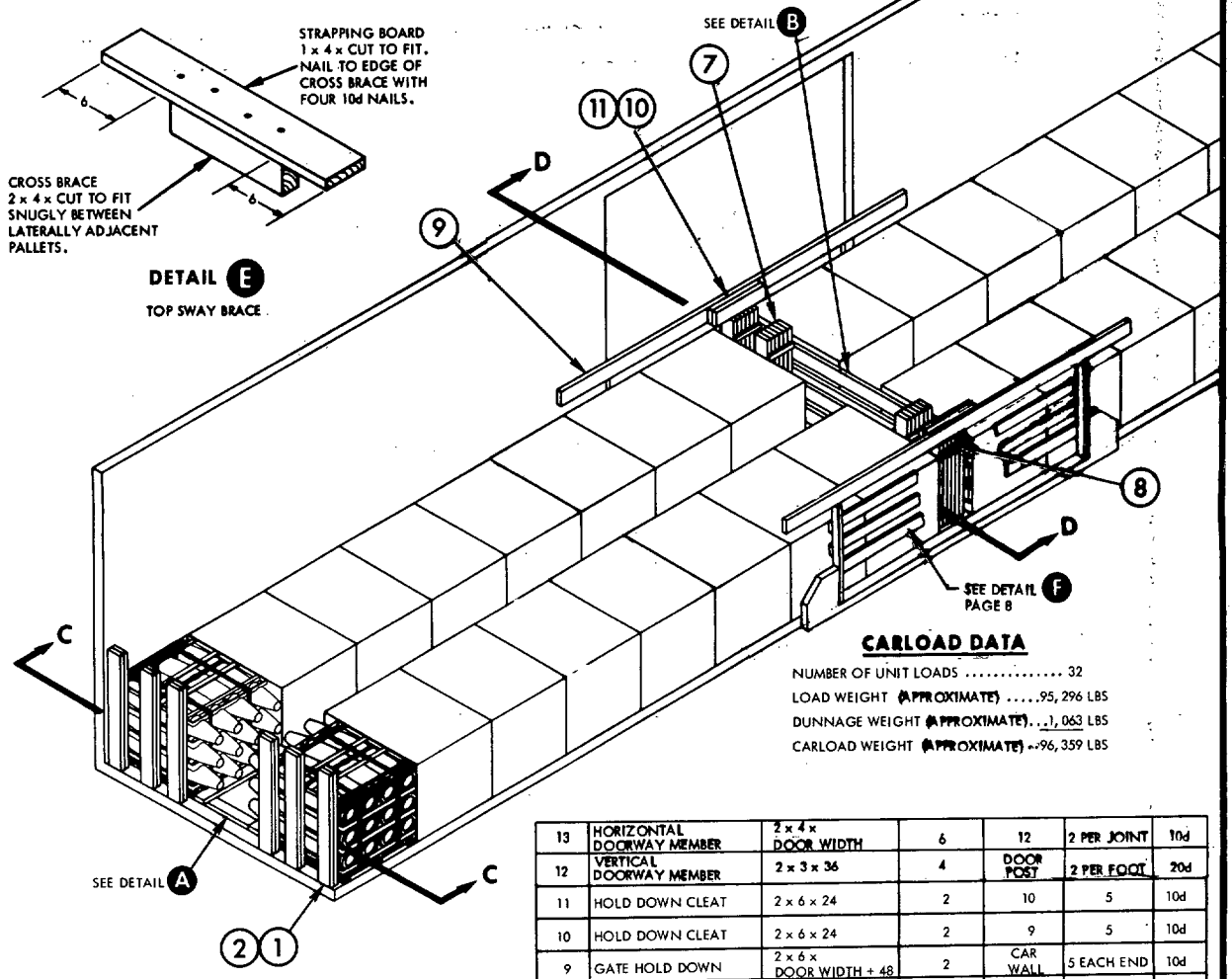


SECTION D-D

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LOADING PLAN



CARLOAD DATA

NUMBER OF UNIT LOADS 32
LOAD WEIGHT (APPROXIMATE) .....95, 296 LBS
DUNNAGE WEIGHT (APPROXIMATE)...1, 063 LBS
CARLOAD WEIGHT (APPROXIMATE) ..96, 359 LBS

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
13	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	6	12	2 PER JOINT	10d
12	VERTICAL DOORWAY MEMBER	2 x 3 x 36	4	DOOR POST	2 PER FOOT	20d
11	HOLD DOWN CLEAT	2 x 6 x 24	2	10	5	10d
10	HOLD DOWN CLEAT	2 x 6 x 24	2	9	5	10d
9	GATE HOLD DOWN	2 x 6 x DOOR WIDTH + 48	2	CAR WALL	5 EACH END	10d
8	CENTER GATE TIE CLEAT	1 x 2 x TO SUIT	8	6, 7	1 PER JOINT	10d
7	CENTER GATE VERTICAL WEDGES	2 x 6 x 37 1/2	AS REQ'D	SEE NOTE **		
6	CENTER GATE VERTICAL	2 x 6 x 37 1/2	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	6	6	3 PER JOINT	10d
4	SWAY BRACE LONGITUDINAL	2 x 4 x 36 3/8	32	3	3 PER JOINT	10d
3	SWAY BRACE LATERAL	2 x 4 x CUT TO FIT	32	CAR FLOOR	SEE NOTE *	
2	END WALL MEMBER	2 x 6 x 48	12	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 48	12	CAR WALL	1 PER FOOT	10d

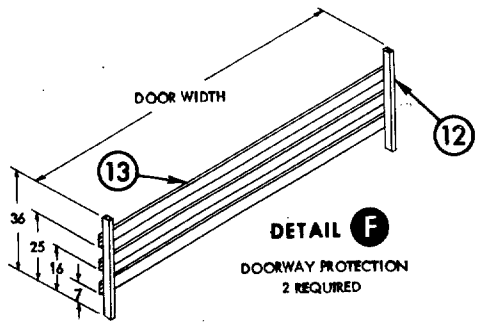
LIST OF MATERIALS AND NAILING DATA

*NAIL LATERAL MEMBERS OF SWAY BRACE TO CAR FLOOR WITH TWO 16d NAILS. EACH END. FOR NAILABLE STEEL FLOOR CARS, WHEN NECESSARY, RELOCATE OR ADD LATERAL MEMBERS TO SWAY BRACE AT NAILABLE LOCATION.

**ADD SUFFICIENT VERTICAL WEDGES FOR A TIGHT FIT. DRIVE IN THE LAST ONE OR TWO AS NECESSARY. 1 x 6 MATERIAL MAY ALSO BE USED AS REQUIRED FOR A TIGHT FIT.

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WHEN BOXCAR HAS STEEL DOOR POSTS
DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY)
PAGE 46, FIG. 21 IS REQUIRED.



DETAIL F
DOORWAY PROTECTION
2 REQUIRED

REVIEW ACTIVITY:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N205)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-STD-1325-9 (NAVY)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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